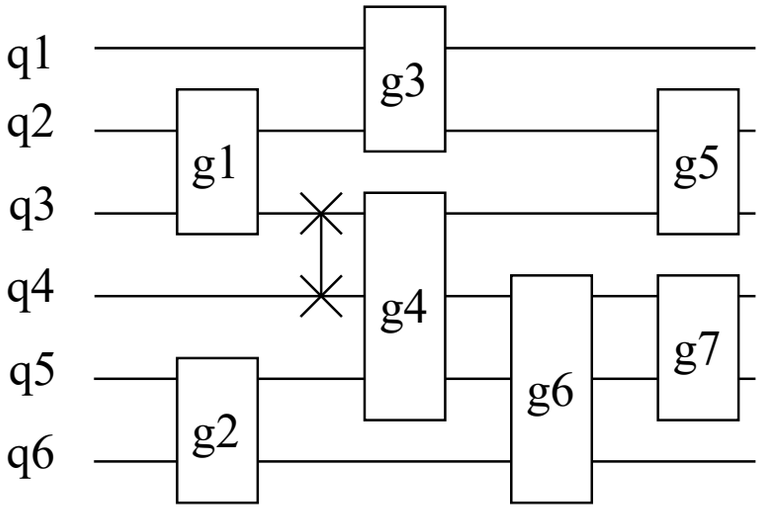
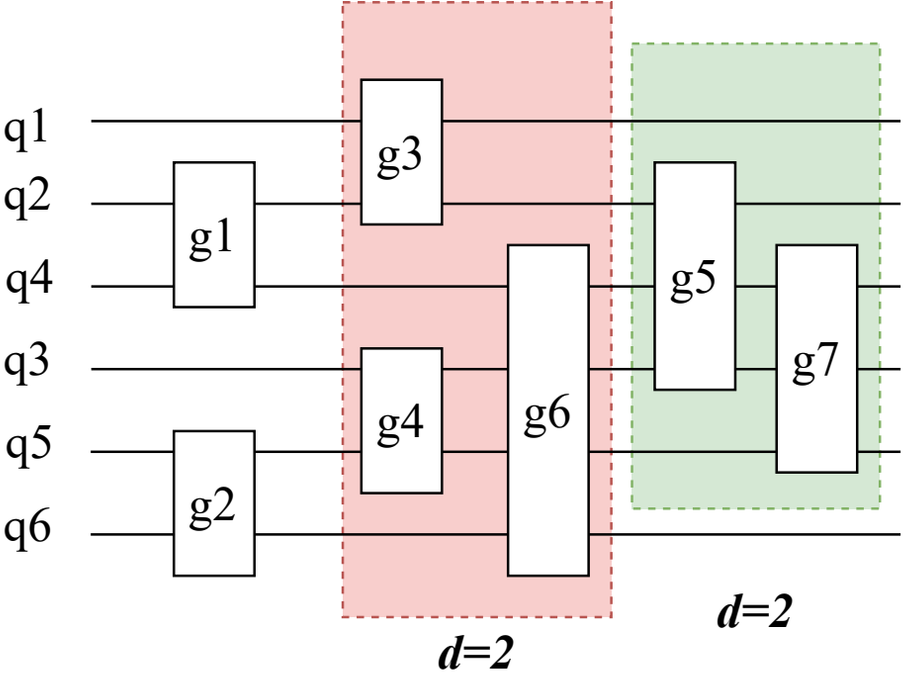
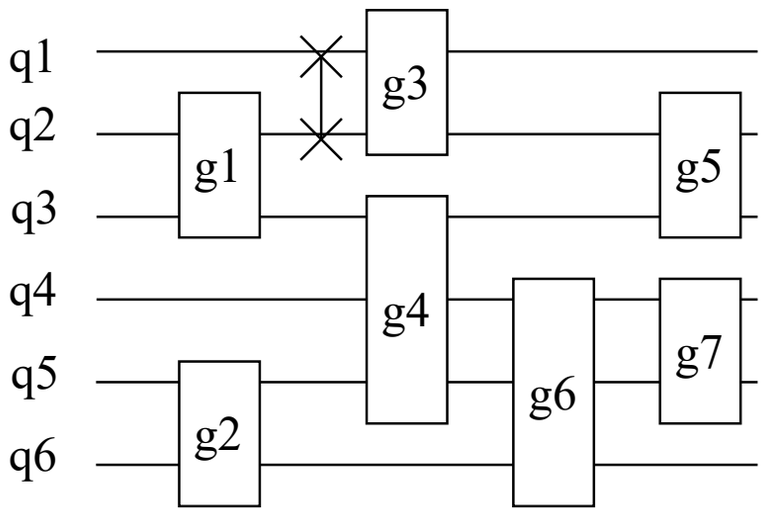


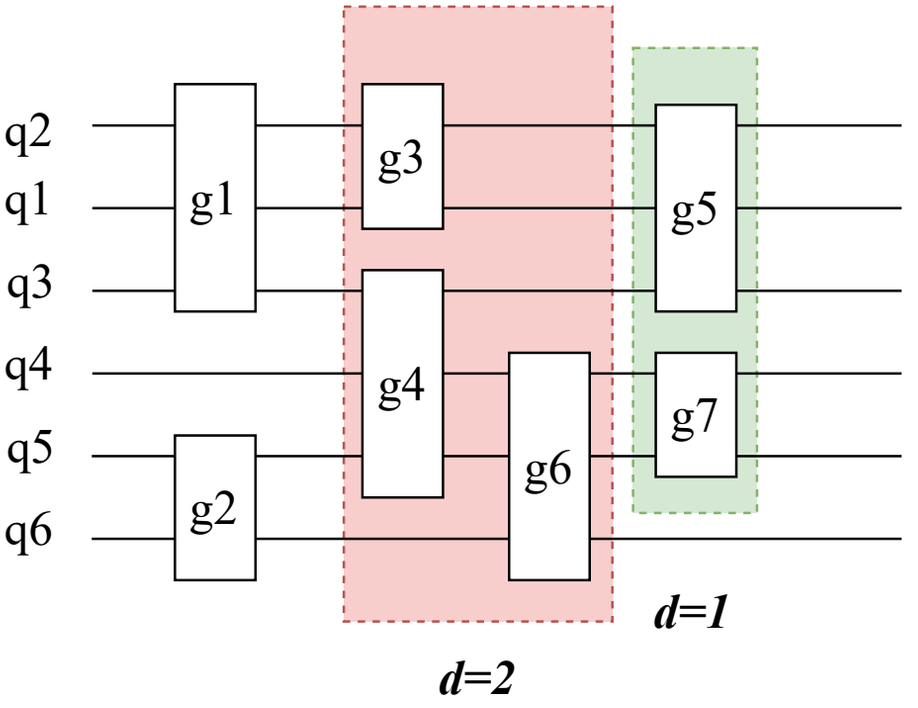
(a) SWAP between q_3 and q_4



(b) Routing cost after SWAP(q_3, q_4)



(c) SWAP between q_1 and q_2



(d) Routing cost after SWAP(q_1, q_2)